

Arjun Mittal

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Experience

Apple | Software Engineer

October 2021 – Present

Input Systems Experience (Swift/Python/C++)

Cupertino, California

- **Architected and developed** the foundational library integrating OpenAI capabilities into the Apple ecosystem, powering key features in Siri and Writing Tools
- **Developed** inline text predictions across all Apple keyboards by integrating predictive model outputs, delivering a **faster and more intelligent** typing experience
- Improved **p95 latency** of autocorrection candidate generation on multilingual keyboards by 30% by **profiling, refactoring, and parallelizing** per-locale computations
- Partnered with **modeling** teams to **productize transformer language models**, enabling seamless autocorrection across Apple keyboards

Wealthsimple | Software Engineer Intern

May 2021 – August 2021

Growth & Data (Python)

Toronto, Canada

- **Optimized** the friend recommendation system by cleaning contact data and improving ranking performance to support real time updates
- **Integrated** Reddit ads API into our ad spend pipeline using Python, Airflow, and DBT automating campaign performance ingestion for unified dashboard reporting

RBC Capital Markets | ETF – Quantitative Trading Analyst

May 2019 – August 2020

ETF Market Making (Python/KBD)

Toronto, Canada

- Developed trading application interface using React & Java, **allowing concurrent traders to make real-time system updates and manage the execution of over 60% of all Canadian ETFs**
- Led the research and development of a factor-based ETF pricing validation tool mitigating erroneously priced orders, leading to **annual savings of a 5-figure amount**
- Devised a strategy to **manage book risk during periods of high market volatility**, reducing costs by a 6-figure amount annually
- Reduced run time of scripts by 80%, improving availability of decision critical reports through **vectorizing query operations** and **migrating codebase** to use the native database language

Scotiabank | Payments - Software Engineering Intern

May 2018 – August 2018

Payments (Java)

Toronto, Canada

- Collaborated on ScotiaConnect, a digital banking platform, creating an automated end-to-end secure file transfer management system, **supporting transactions valued at more than \$3 billion** as of 2018
- **Architected real-time dashboard on Google Cloud** to help senior executives monitor customer KPIs

Technical Skills

Languages | C++ • Swift • Python • Objective-C • C • SQL • KDB

Education

University of Toronto

2021

Bachelor of Applied Science, Computer Engineering

Toronto, Canada

Coursework - Machine Learning, Algorithms & Data Structures, Operating/Distributed Systems, Computer Networks

Projects

CockroachDB – Open-Source | Go, C++

2021

Contributed to CockroachDB, a distributed SQL database, expanding its geospatial capabilities with new functions

Facebook Libra – Smart Contract Verification | Go, React, JavaScript, AWS, Docker

2021

Engineered a system for the automatic verification and synthesis of Facebook's Libra smart-contracts